COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S1
DATE PERFORMED:	7-10-2026	**DATE SUBMITTED:**	7-14-2026
NAME:	GUARIN, JAN MIKAELA C.	**INSTRUCTOR:**	ENGR. JIMLORD QUEJADO

ACTIVITY NO. 1.1 BASIC C++ PROGRAMMING

1. PROCEDURE OUTPUT

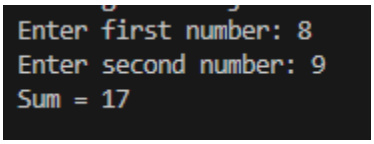
ILO A.

1. Create a function that will take two numbers and display the sum

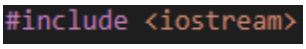
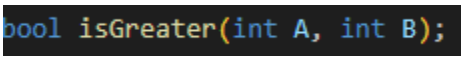
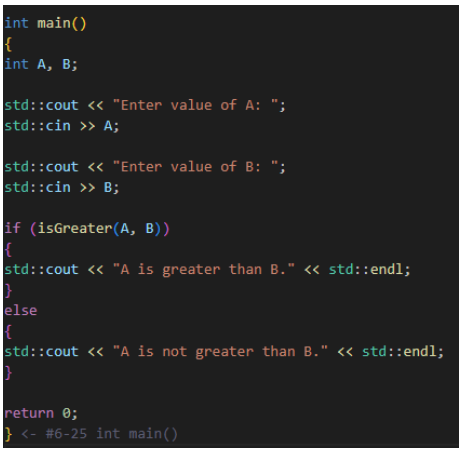
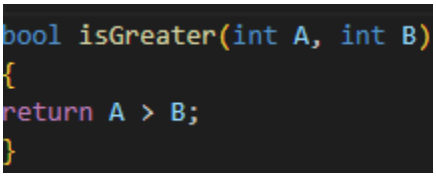
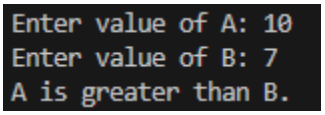
Sections	Answer
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	`void displaySum(int num1, int num2);`
Class Declaration and Method Definition Section	N/A

Main Function	`int main()`
	`{`
	`int num1, num2;`
	
	`std::cout << "Enter first number: ";`
	`std::cin >> num1;`
	
	`std::cout << "Enter second number: ";`
	`std::cin >> num2;`
	
	`displaySum(num1, num2);`
	
	`return 0;`
	`} <- #6-18 int main()`

Method Definition	`void displaySum(int num1, int num2)`
	`{`
	`std::cout << "Sum = " << num1 + num2 << std::endl;`
	`}`

Output	

2. Create a function that will return whether variable A is greater than variable B.

Sections	**Answer**
Header File Declaration Section	
Global Declaration Section	
Class Declaration and Method Definition Section	N/A
Main Function	
Method Definition	
Output	

3. Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.

Sections	Answer
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	`bool logicalOperations(bool A, bool B);`
Class Declaration and Method Definition Section	N/A

Main Function	`int main()`
	`{`
	`bool A, B;`
	
	`std::cout << "Enter value for A (1=True, 0=False): ";`
	`std::cin >> A;`
	
	`std::cout << "Enter value for B (1=True, 0=False): ";`
	`std::cin >> B;`
	
	`if (logicalOperations(A, B))`
	`{`
	`std::cout << "Function executed successfully." << std::endl;`
	`}`
	
	`return 0;`
	`} <- #6-21 int main()`

Method Definition	`bool logicalOperations(bool A, bool B)`
	`{`
	`std::cout << "\nLogical Operations\n";`
	`std::cout << "A AND B = " << (A && B) << std::endl;`
	`std::cout << "A OR B = " << (A || B) << std::endl;`
	`std::cout << "NOT A = " << (!A) << std::endl;`
	`std::cout << "NOT B = " << (!B) << std::endl;`
	
	`return true;`
	`} <- #24-32 bool logicalOperations(bool A, bool B)`

Output	`Enter value for A (1=True, 0=False): 1`
	`Enter value for B (1=True, 0=False): 0`
	
	`Logical Operations`
	`A AND B = 0`
	`A OR B = 1`
	`NOT A = 0`
	`NOT B = 1`
	`Function executed successfully.`

ILO B.

TRIANGLE

```cpp
#include <iostream>

class Triangle{
    private:
        double totalAngle, angleA, angleB, angleC;

    public:
        Triangle(double A, double B, double C);
        void setAngles(double A, double B, double C);
        const bool validateTriangle();
};

Triangle::Triangle(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

void Triangle :: setAngles (double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}
```

```cpp
const bool Triangle::validateTriangle()  {
return (totalAngle == 180);

}

int main(){
    Triangle set1(40, 30, 110);
        if(set1.validateTriangle()){
            std::cout << "The shape is a valid triangle.\n";
        } else {
            std::cout << "The shape is NOT a valid triangle.\n";
        }
        return 0;
}
```

Figure1. Code Snippet of Triangle

```
c:\Users\mikaela\Downloads\GUARIN_CPE21S2\ACTIVITY 1\output>.\"TRIANGLE.exe"
The shape is a valid triangle.
```

Figure2. Output of Triangle

ANALYSIS:

In this program, I used the **class** keyword to create the **Triangle** class, **private** and **public** access specifiers to control data access, a constructor (**Triangle()**) to initialize the object's values, and member functions (**setAngles()** and **validateTriangle()**) to perform specific tasks. I also used the **bool** data type for the validation function, the return statement to send back the result, an **if-else** statement for decision-making, **std::cout** to display the output, and created an object (**Triangle set1**) to access the class functions.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. Create a C++ program to swap the two numbers in different variables.

```cpp
#include <iostream>

void swapNumbers(int &a, int &b)
{
    int temp = a;
    a = b;
    b = temp;
}

int main (){
    int num1, num2;
    std::cout << "Enter the first number: ";
    std::cin>>num1;

    std::cout << "Enter second number: ";
    std::cin>>num2;

    std::cout << "\nBefore Swapping\n";
    std::cout << "First Number: " << num1 << std:: endl;
    std::cout << "Second Number: " << num2 << std:: endl;

    swapNumbers(num1,num2);
    std::cout << "\nAfter Swapping\n";
    std::cout << "First Number: "<< num1 << std:: endl;
    std::cout << "Second Number: " << num2 << std:: endl;

    return 0;
}
```

Figure1. Code Snippet of Number Swapping

```
c:\Users\mikaela\Downloads\GUARIN_CPE21S2\ACTIVITY 1\output>.\"SUPPLEMENTARY1.exe"
Enter the first number: 18
Enter second number: 10

Before Swapping
First Number: 18
Second Number: 10

After Swapping
First Number: 10
Second Number: 18
```

Figure2. Output of Number Swapping

ANALYSIS:

This program swaps the values of two user-entered numbers using a separate function called swapNumbers(). The function uses **reference parameters (&)**, which allows it to change the original values of the variables without returning anything. A temporary variable named temp is used to store one of the values while the swap takes place. The program first displays the numbers before swapping, calls the function, and then displays the updated values after the swap.

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

```cpp
#include <iostream>

double kelvintoFahrenheit(double kelvin)
{
    return (kelvin - 273.15) * 9/5+32;
}

int main ()
{
    double kelvin;

    std::cout << "Enter temperature in Kelvin: ";
    std::cin >> kelvin;

    std:: cout << "Temperature in Fahrenheit: "
               << kelvintoFahrenheit(kelvin)
               << std:: endl;

return 0;

}
```

Figure3. Code Snippet of Kelvin to Fahrenheit

```
c:\Users\mikaela\Downloads\GUARIN_CPE21S2\ACTIVITY 1\output>.\"SUPPLEMENTARY2.exe"
Enter temperature in Kelvin: 79
Temperature in Fahrenheit: -317.47
```

Figure4. Output of Kelvin to Fahrenheit

ANALYSIS:

This program converts a temperature from **Kelvin to Fahrenheit** using a user-defined function called **kelvintoFahrenheit()**. The function receives the temperature in Kelvin, applies the conversion formula, and returns the equivalent temperature in Fahrenheit. In the **main()** function, the user enters a temperature in Kelvin, which is then passed to the conversion function. The converted value is displayed using **std::cout.**

3. Create a C++ program that has a function that will calculate the distance between two points.

```cpp
#include <iostream>
#include <cmath>
using namespace std;

double distance(double x1, double y1, double x2, double y2)
{
    return sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));
}

int main()
{
    double x1, y1, x2, y2;

    cout << "First point (x y): ";
    cin >> x1 >> y1;

    cout << "Second point (x y): ";
    cin >> x2 >> y2;

    cout << "\nDistance = " << distance(x1, y1, x2, y2);

    return 0;
}
```

Figure5. Code Snippet of Distance between Two Points

```
c:\Users\mikaela\Downloads\GUARIN_CPE21S2\ACTIVITY 1\output>.\"SUPPLEMENTARY3.exe"
First point (x y): 7 10
Second point (x y): 9 18

Distance = 8.24621
```

Figure6. Output of Distance between Two Points

ANALYSIS:

This program calculates the distance between two points using a function called **distance()**. The function uses the distance formula with **sqrt()** and **pow()** from the cmath library to get the result. In the **main()** function, the user inputs the coordinates of two points, which are passed to the function to calculate the distance.

4. Create a C++ program that has a function that will calculate the distance between two points. Modify the code given in ILO B and add the following functions:
    a. A function to compute for the area of a triangle
    b. A function to compute for the perimeter of a triangle]

c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

```cpp
#include <iostream>
#include <cmath>
using namespace std;

double area(double a, double b, double c)
{
    double s = (a + b + c) / 2;
    return sqrt(s * (s - a) * (s - b) * (s - c));
}

double perimeter(double a, double b, double c)
{
    return a + b + c;
}

string triangleType(double a, double b, double c)
{
    double x = a;
    double y = b;
    double z = c;

    if (x > y)
        swap(x, y);
    if (y > z)
        swap(y, z);
    if (x > y)
        swap(x, y);

    if (x + y <= z)
        return "Others (Invalid Triangle)";

    double left = (x * x) + (y * y);
    double right = z * z;

    if (left > right)
        return "Acute Triangle";
    else if (left < right)
        return "Obtuse Triangle";
    else
        return "Others (Right Triangle)";
}

int main()
{
    double side1, side2, side3;

    cout << "Enter side 1: ";
    cin >> side1;

    cout << "Enter side 2: ";
    cin >> side2;

    cout << "Enter side 3: ";
    cin >> side3;

    if (side1 + side2 > side3 &&
        side1 + side3 > side2 &&
        side2 + side3 > side1)
    {
        cout << "\nArea = " << area(side1, side2, side3) << endl;
        cout << "Perimeter = " << perimeter(side1, side2, side3) << endl;
        cout << "Triangle Type = " << triangleType(side1, side2, side3) << endl;
    }
    else
    {
        cout << "\nThe given sides do not form a triangle." << endl;
    }

    return 0;
}
```

Figure7. Code Snippet of Area, Perimeter, Type of a Triangle

```
c:\Users\mikaela\Downloads\GUARIN_CPE21S2\ACTIVITY 1\output>.\"SUPPLEMENTARY4.exe"
Enter side 1: 8
Enter side 2: 10
Enter side 3: 7

Area = 27.8107
Perimeter = 25
Triangle Type = Acute Triangle
```

Figure8. Output of Area, Perimeter, Type of a Triangle

ANALYSIS:

This program calculates the **area, perimeter, and type of a triangle** given three side lengths. It uses **#include <cmath>** for the **sqrt()** function needed in calculating the area. The program uses **double** to store decimal values,

string for the triangle classification, and **if-else** statements to check conditions and determine the triangle type. I used functions such as **area(), perimeter()**, and **triangleType()** to organize the calculations, while **return** is used to provide results from each function. The program applies **Heron's Formula** to calculate the area and the **Pythagorean theorem** to identify whether the triangle is acute, obtuse, or right-angled. It also uses **swap()** to rearrange the side values and checks whether the given sides form a valid triangle before displaying the results.

SUMMARY OF LESSONS LEARNED

In this activity, I learned how to apply different C++ concepts such as functions, classes, objects, conditional statements, and mathematical operations. I realized that using functions makes a program more organized and easier to understand because each function has a specific task. I also learned how classes and objects work in C++ and how they can be used to structure a program better. Through the supplementary activities, I was able to practice solving different problems like swapping numbers, converting temperatures, finding distances, and calculating triangle measurements.

CONCLUDING STATEMENT/FEEDBACK

I think I performed well in this activity because I was able to complete the programs and understand the purpose of each concept used. I learned that proper organization of code makes programs easier to create and troubleshoot. However, I still need to improve my debugging skills and practice writing more complex programs to become more confident in C++.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Chatgpt**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Code Assistance**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.


T.I.P
TECHNOLOGICAL INSTITUTE OF THE PHILIPPINES
COMPUTER ENGINEERING
QUEZON CITY